The allergy epidemic extends beyond the past few decades.

. . . . .

BACKGROUND: Increased prevalence of allergic diseases in western societies has been described as an epidemic. The precise turning point for the epidemic and the antigens responsible for it remain obscure. OBJECTIVE: To evaluate how the prevalence of atopic disease has changed in terms of detectable sensitization to aeroallergens and dietary allergens in a cross-sectional comparison of subjects from birth cohorts more than 60 years apart. METHODS: We studied four groups of 100 subjects each (at ages 7, 27, 47 and 67 years), representing those born in 1990, 1963-66, 1943-46 and in 1923-26, respectively. Serum total and specific IgE concentrations against aeroallergens and dietary allergens were determined. A questionnaire elicited information on symptoms, allergic diseases and medication. RESULTS: The proportion of subjects with detectable IgE antibodies against aeroallergens increased consistently from the oldest to the youngest birth cohorts; chi2 trend=56.809, $P<0.0001$. Similar progression was not seen in sensitization to dietary allergens. The proportion of those with diagnosed asthma differed significantly (chi2=13.45, $P=0.004$) across the birth cohorts. The lowest prevalence of asthma and sensitization to dietary allergens was detected in those born in 1943-46, i.e. during or immediately after World War II. CONCLUSION: Prevalence of sensitization to airborne allergens, unlike that to dietary allergens, has increased over a long period of time. Our results support the concept of the immune function being programmed by external factors early in life. They also call for caution when interpretations of the pace and possible causes of the allergy epidemic are made on the basis of short-term studies.